PRISM: Background and Related Work on Privacy-Constrained Customer Churn Classification

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Abstract

Churn prediction helps companies decide who to try to keep, but a useful model might have to work with only a small amount of customer information. We looked at how churn modelling developed, went over predictive methods and data-minimization methods, and compared what data each one needs and where each falls short. PRISM looks at what we lose in predictive performance when account attributes are restricted into fixed visibility tiers. What we contribute is a controlled comparison across different levels of information availability. We are not claiming this guarantees privacy. Repository: https://github.com/ParsanaDev/PRISM-churn-privacy.

1 Introduction

Churn just means losing existing customers. For subscription services, recorded cancellations give us a natural baseline, and they become the labels for a supervised binary classification problem (churn or no churn). The question PRISM asks is whether we can still make useful predictions when only a limited set of account attributes is available. Earlier work tests predictive methods and looks at how to maximize profit from retention [4, 5]. Separately, data-minimization research asks how much input information a model actually needs [3]. PRISM's main comparison comes from putting these two together: we measure how churn prediction performance changes as we limit what the model can see, using clearly defined attribute tiers. We only treat this as a question of predictive cost. Account metadata can still be personal data, so using fewer attributes doesn't by itself mean privacy is protected or that a legal requirement is met.

2 Historical Background

The underlying problem is older than computers. Copeland's 1923 discussion of buying habits and brand preference or insistence [2] already explains why buyers keep going back to a seller or switch to another one. By 2000, Mozer et al. [4] were applying statistical machine learning to churn among wireless subscribers, using operational customer records. After that, other work compared many modelling choices to find out which ones work best [5]. Scalable tree boosting [1] added to the available tools, and input-minimization methods [3] moved the focus toward keeping a model useful while giving it less detailed information.

3 Hierarchy of Related Approaches

Figure 1 sorts the methods we reviewed by what they mainly do.

Branch	Methods
Predictive methods	Statistical: logistic regression [4]. Tree-based: single or boosted [1, 4]. Neural: multilayer networks [4].
Input minimization	Feature suppression [3]. Value generalization [3].

Figure 1: *Two-level hierarchy (churn modelling and input minimization) by methodological function. Input restrictions can be combined with each predictive family.*

Predictive methods estimate churn from labelled customer records, and their subcategories separate statistical, tree-based and neural models. Input-minimization methods change the information given to a classifier (the predictor), either by suppressing features or by generalizing values. The two branches are independent, so any restricted feature set can be paired with any classifier family. If we varied both at once we couldn't tell their effects apart. So PRISM fixes the classifier and changes the information tier, and classifiers are only compared against each other inside one tier. The hierarchy only includes methods that matter for this comparison.

4 Comparative Analysis

Statistical and neural classifiers. Mozer et al. [4] compare logistic regression, decision trees and neural networks on subscriber records covering usage, billing and complaints. Logistic regression is an easy-to-interpret baseline, but it treats each attribute's effect separately and misses interactions unless someone adds them by hand. Neural networks pick up those interactions automatically, but you pay for it in tuning effort and in being harder to interpret. One important point: neither family needs behavioural logs by design. That dependence comes from the inputs we give them. So if we remove logs, we have to retrain and measure again. It doesn't mean the classifier stops being usable.

Tree-based methods and evaluation. A decision tree splits customers with a series of attribute tests. The rules are easy to read, but the partitions are unstable and the tree overfits easily. XGBoost [1] deals with this by building trees one after another under a regularized objective, with each tree correcting the errors of the earlier ones. That makes it a strong candidate for tabular account data, though it's harder to read and takes more tuning. Good results on general benchmarks don't prove it will win under PRISM's restricted tiers, because those benchmarks assume full access to features. Neslin et al. [5] compare many submitted churn models on shared data and find that the modelling approach, and not just the algorithm, drives predictive accuracy. They also turn that accuracy into profit for a retention campaign. That speaks to business value, but it depends on economic assumptions and still doesn't say which attributes can be withheld. So the choice of algorithm, the evaluation objective and the permitted information are three separate decisions.

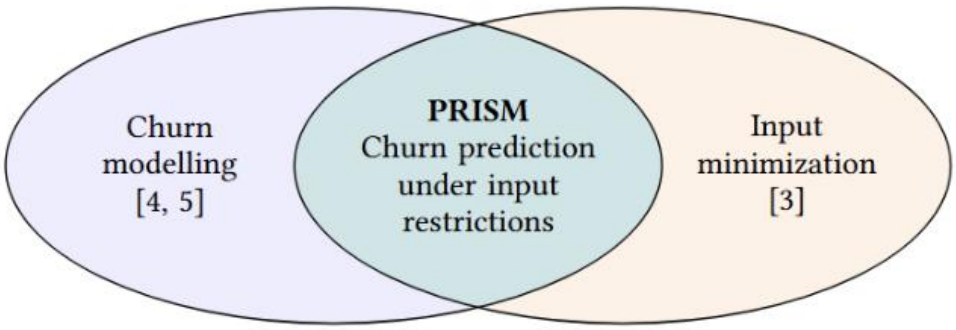


Figure 2: *Conceptual overlap of research aims. The placement shows what this review emphasizes. It does not mean exclusive membership or that the intersection is unstudied.*

Table 1: *Representative approaches. The limitations are about how suitable each approach is for PRISM. They are qualitative comparisons and not experimental results.*

Approach	Input / mechanism	Strength	Limitation
Logistic [4]	Encoded records; weighted log-odds	Simple baseline	Interactions need design
Single tree [4]	Attribute splits	Readable rules	Unstable partitions
XGBoost [1]	Tabular features; boosted trees	Nonlinear interactions	Tuning; complex ensemble
Neural network [4]	Encoded records; learned layers	Flexible mapping	Training and interpretation
Selection [6]	Relevance scores / feature elimination	Focuses predictors	No fixed access restriction
Input reduction [3]	Suppress or generalize inputs	Privacy control	Existing model required

Feature selection versus information restrictions. Omari et al. [6] use mutual information and recursive feature elimination with logistic regression to pick useful features from the IBM Telco data. These methods choose features by how much they help predict churn. In PRISM we decide what information each tier allows before we evaluate any model. A feature can be very useful and still be unavailable under a given restriction. The study also uses a Churn Score, so we'd have to check how it was built and whether it would actually exist at prediction time. Feature selection by itself doesn't tell us how a model does when some information can't be collected at all.

Input minimization. Goldsteen et al. [3] cut down input information by removing features or making their values less precise, while keeping the loss in accuracy small. They use a surrogate decision tree, which is a simpler model that approximates what an existing model predicts, to decide how to simplify the inputs. The original model isn't retrained. Suppression removes a feature. Generalization swaps an exact value for a less precise one, like a range. Their approach needs a trained model and suitable data. In PRISM, we'll train and test each classifier using only the information allowed in its tier. Table 1 compares the approaches, and Figure 2 shows how PRISM links churn prediction with input minimization.

5 Positioning PRISM

PRISM studies how restricting customer information changes churn prediction. We propose three information tiers: basic administrative attributes; administrative plus billing attributes; and all eligible account and service attributes. Each tier includes everything from the one before it. We'll check the dataset before assigning features to tiers, and we'll leave out identifiers and any fields derived from the churn outcome. We'll compare logistic regression and boosted trees and retrain each model for every tier. All experiments will use the same data splits and similar tuning budgets, and preprocessing will be fitted on training data only.

We'll report PR-AUC, ROC-AUC and churn recall, with uncertainty estimates. We'll also measure recall within a fixed proportion of customers ranked as most likely to leave, and we'll pick that proportion before evaluating any models. Feature counts will describe how much information each tier uses, but they don't measure privacy protection. If the dataset has subscription details but no activity logs, our findings will only apply to restrictions on account information. Testing performance on new customers will need

its own evaluation on customers with short account histories. Overall, this benchmark tests how much predictive performance changes as more account information becomes available.

References

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